explAIn

How we got here

Almost none of the maths is new. Compute, data and one architecture changed.

"Almost none of the maths is new."

Ask on every project

1. Which part of this is actually new: the maths, the architecture, the data or the compute?
2. What is the oldest idea this depends on, and what year is it from?
3. If the maths is old, what changed recently to make it work at all?
4. Has this approach been tried and abandoned before, and what killed it then?
5. Are we promising something sized for hardware and data we do not have yet?

Rules of thumb

- Date the idea before you call it a breakthrough.
- Separate the four layers: mathematics, architecture, data, compute. Novelty sits in one.
- Training is gradient descent on a fitted curve: Cauchy 1847, at industrial scale.
- When a field over-promises, the next winter is a funding decision, not a maths one.

Red flags

- ⚠ A vendor calls a technique new and cannot name a year for it
- ⚠ A roadmap that needs compute or data nobody has yet
- ⚠ "Neural network" used as proof that something is modern
- ⚠ A claim of general intelligence with one demo behind it

Key terms

Gradient descent	Walking downhill on an error curve to find better parameters. Cauchy, 1847.
Backpropagation	The chain rule applied backwards through layers to assign blame. Published 1970.
Perceptron	Rosenblatt's 1958 learning machine: the first neural network you could train.
AI winter	A spell when funding and interest collapsed: roughly 1974–80 and 1987–93.
Transformer	The 2017 architecture that made training parallel across a whole sequence.

CHECKLIST

Date the claim in four questions

- ☐ Write down what the vendor or the paper says is new, in one sentence.
- ☐ Sort it into one of four layers: mathematics, architecture, data, compute.
- ☐ Find the oldest idea it depends on, and the year that idea was published.
- ☐ Ask what changed recently that made the old idea work: usually scale, not insight.
- ☐ If nothing changed except the name, treat it as packaging and price it that way.